

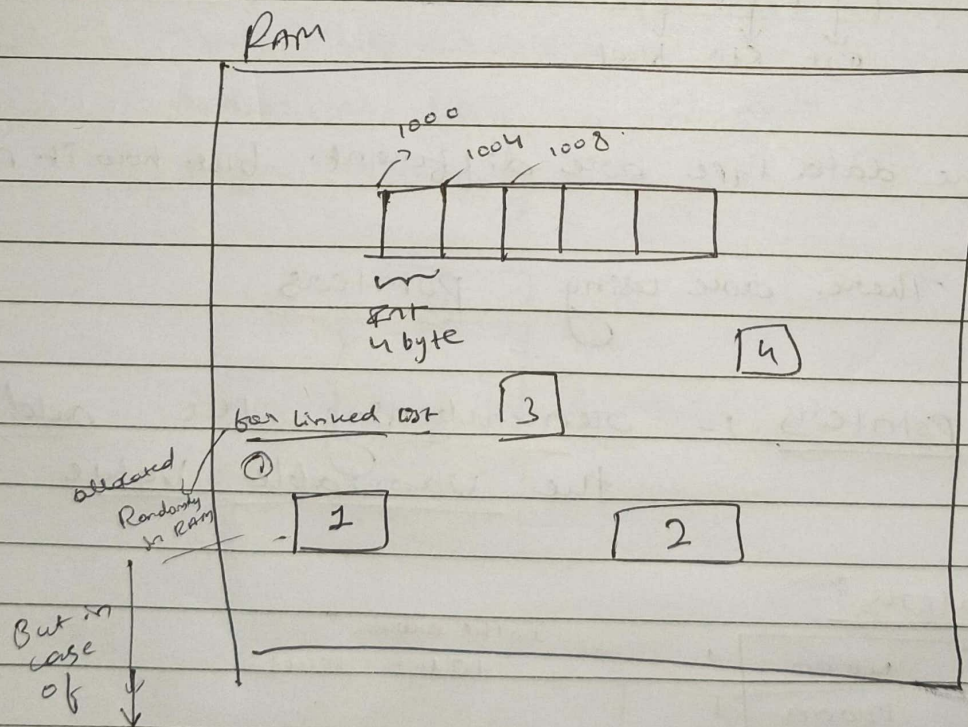
Introduction to Singly linked list

day 29

B Singly linked list in python

Prerequisites:- Object Oriented programming.

- 1) What are linked list?
- 2) What are the disadvantages of arrays that linked list try to overcome?
- 3) Why we need linked list? What are the advantages?
- 4) What are the different operations of linked list?
- 5) How to master writing code for these operations?
- 6) Hands on workshop for writing code for the operations?



In python & Java it get allocated through Objects :-

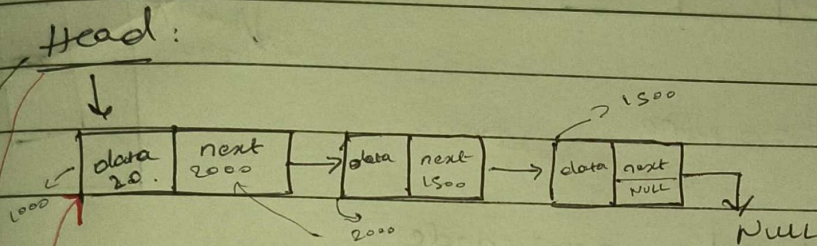
↳ Heap :- The memory allocated here.

Node :-

L7 Data

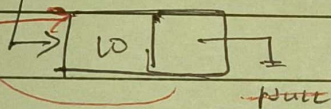
↳ Refrence, Address Next Node

for doubly $\rightarrow$ " " " power node



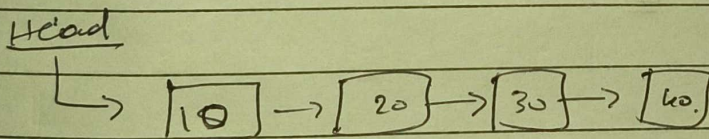
Head $\rightarrow$ Null

new node.next = head



If you the head is not null then we add this to head.

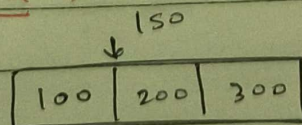
Then ~~test~~ LL look like this. (for odd starting).



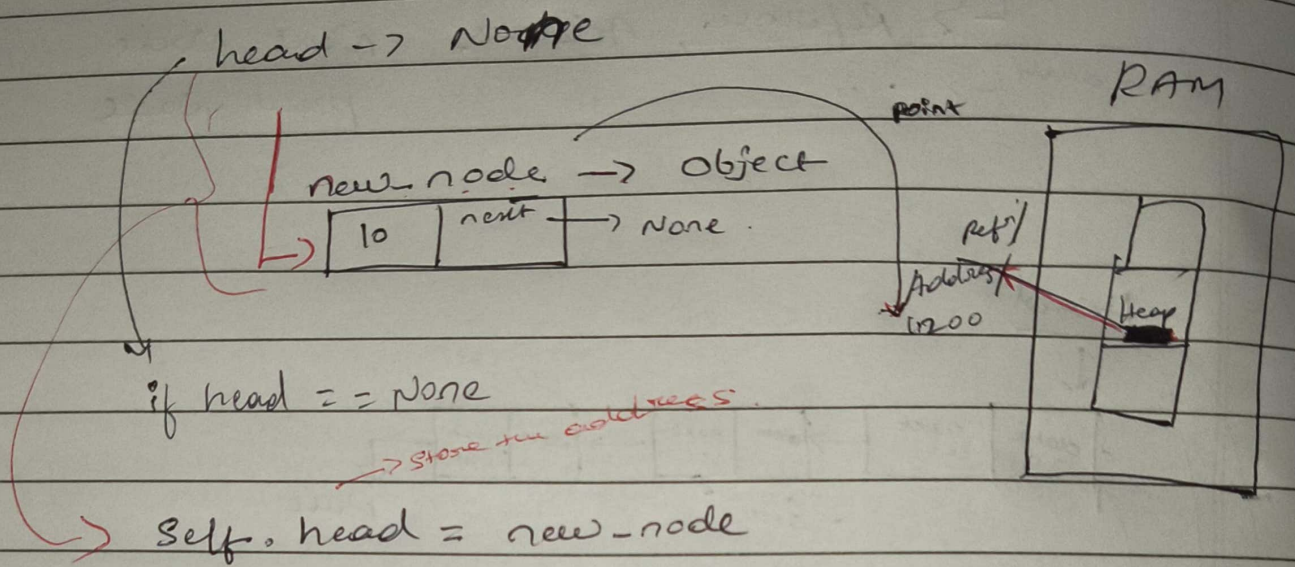
for Enolding to attach end

→ loop (strutting then add last).

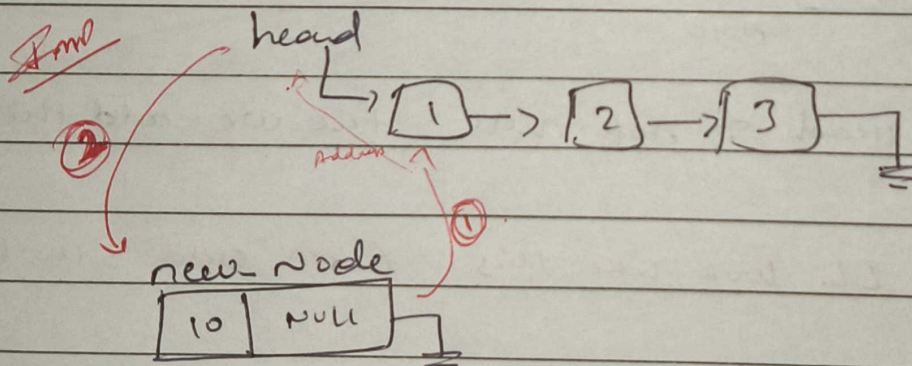
for insert : - (add in b/w)



def insert_at_beginning (data):



head not pointing to Null



new_node = self.head

self.head = new_node

✓ right way

singlylinkedlist.py

SinglyLinkedList.py

06_Data Structure in Python > PracticeSLL > SinglyLinkedList.py > SinglyLinkedList > print_list

```
1 class Node:                                     # class which represent the node in singly linked list
2     def __init__(self, data):
3
4         self.data = data                         # Store data
5         self.next = None                        # Initialize next as None
6
7 class SinglyLinkedList:
8     def __init__(self):
9         self.head = None                        # head is the first node in the linked list
10
11     def insert_at_beginning(self, data):         # creating new node
12         new_node = Node(data)                  # newNode --> data | next
13         new_node.next = self.head              # next --> head
14         self.head = new_node                   # head --> newNode
15
16
17     def print_list(self):                       # start from head/currently in head node
18         current = self.head                     # current --> head
19
20         while current:                          # while current is not None
21             print(current.data, end=" -> ")    # print current node data ex: 10 ->
22             current = current.next              # move to the next node
23             print("None")                      # print None at the end of the list ex: 10 -> 20 -> None
24
25
26 # driver code to test the above class
27 if __name__ == "__main__":
28     sll = SinglyLinkedList()
29
30     # Insert at beginning
31     sll.insert_at_beginning(10) # list: 10 -> None
32     sll.insert_at_beginning(20) # list: 20 -> 10 -> None
33     sll.print_list()           # print the list
```